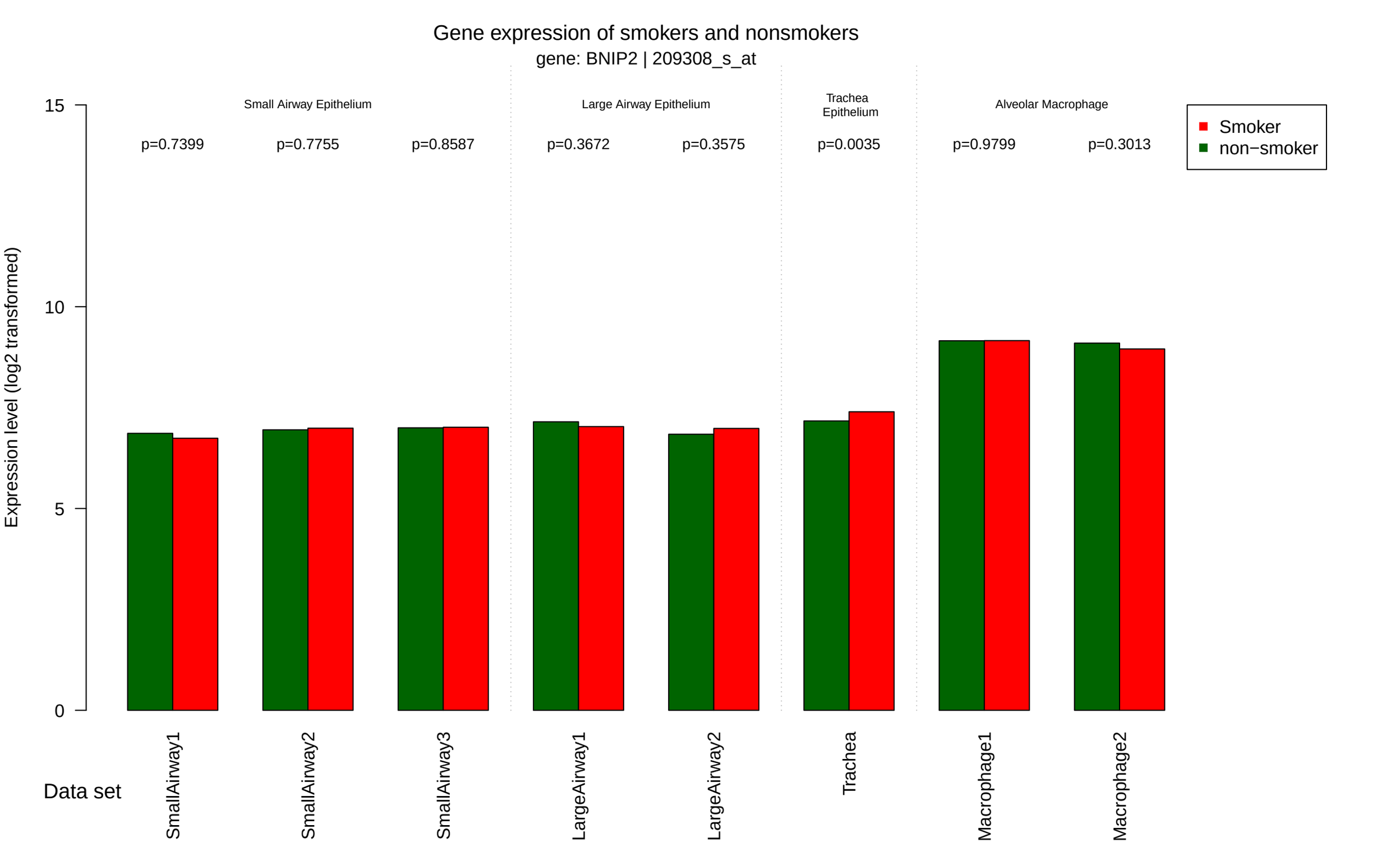
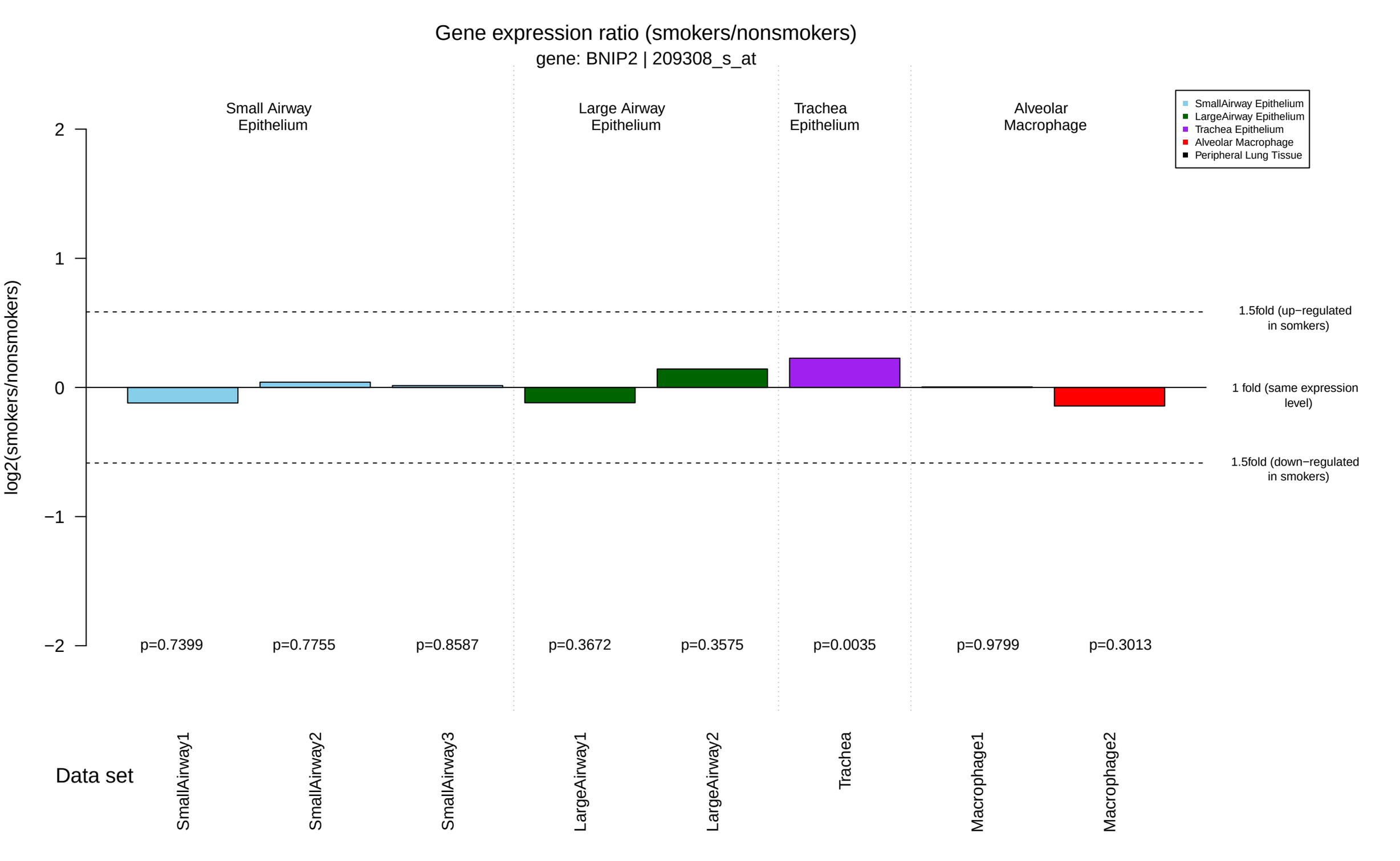
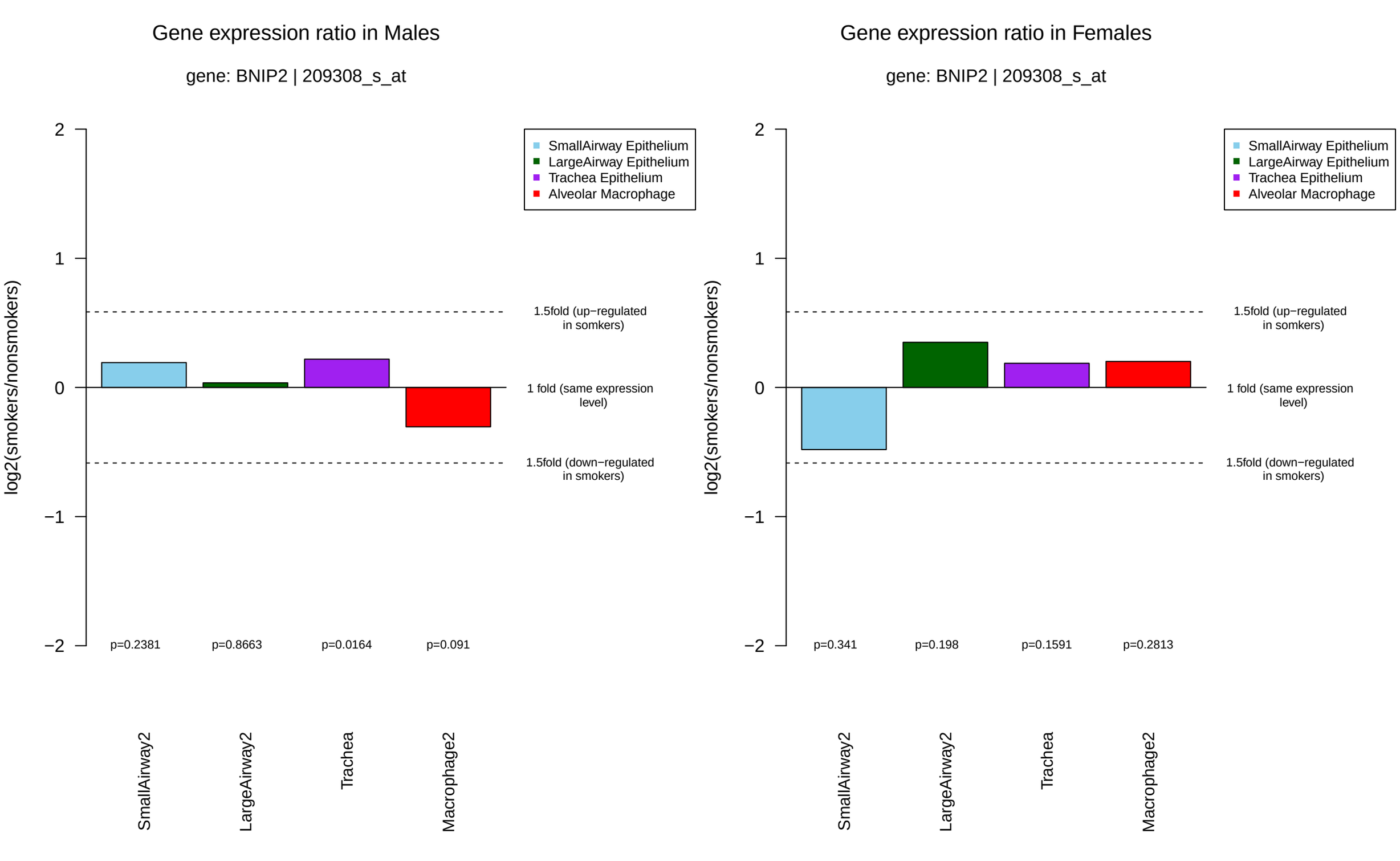
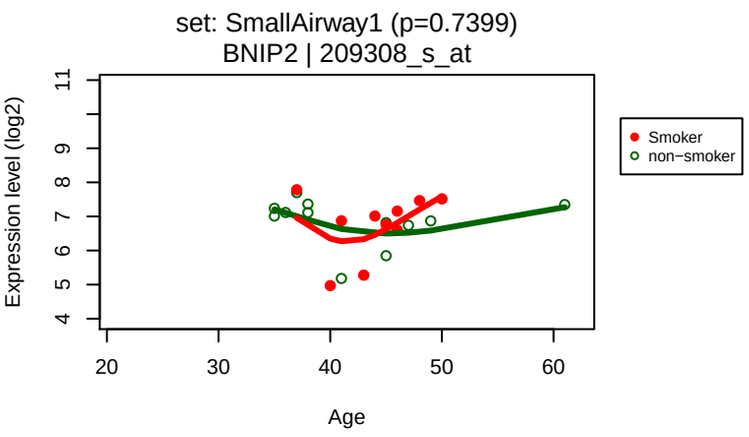
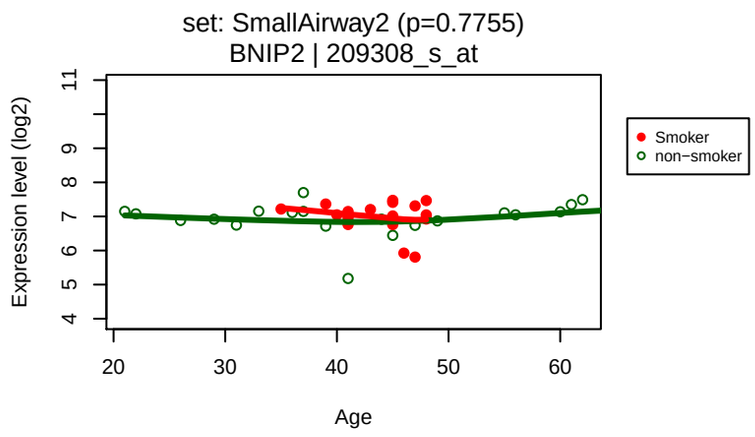
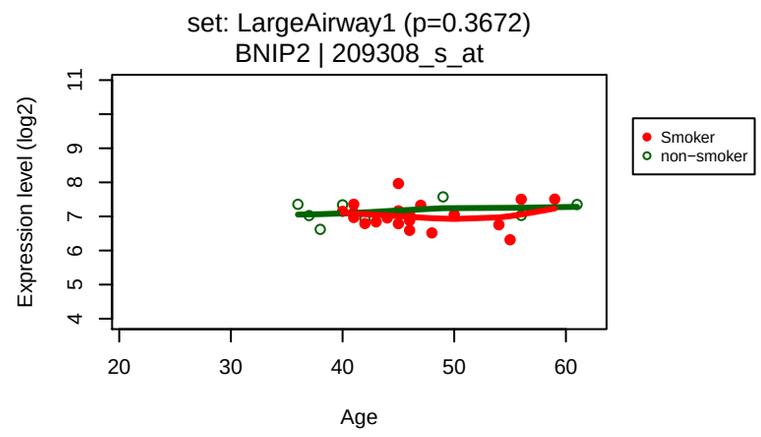

Gene expression of smokers and nonsmokers
gene: BNIP2 | 209308_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7399
p=0.7755
p=0.8587
p=0.3672
p=0.3575
p=0.0035
p=0.9799
p=0.3013
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

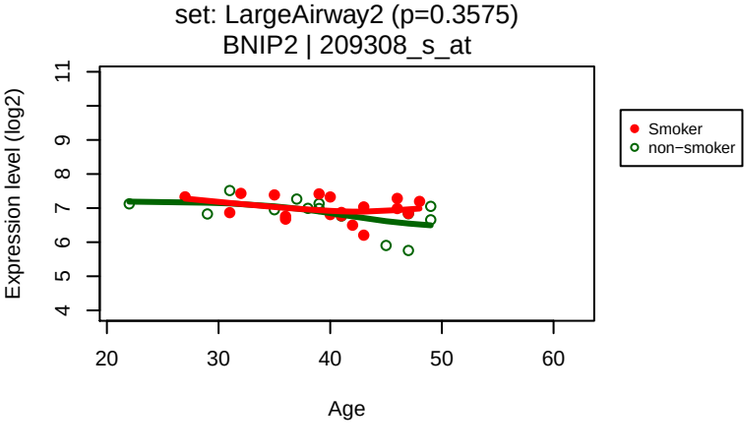

Gene expression ratio (smokers/nonsmokers)
gene: BNIP2 | 209308_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7399
p=0.7755
p=0.8587
p=0.3672
p=0.3575
p=0.0035
p=0.9799
p=0.3013
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

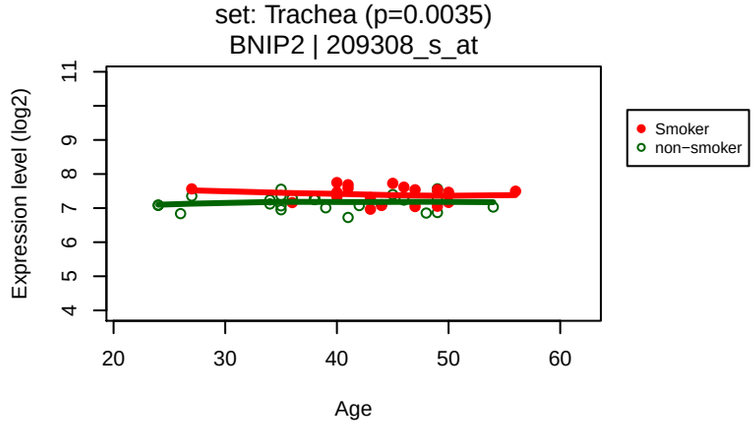

Gene expression ratio in Males
gene: BNIP2 | 209308_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2381
p=0.8663
p=0.0164
p=0.091
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BNIP2 | 209308_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.341
p=0.198
p=0.1591
p=0.2813
SmallAirway2
LargeAirway2
Trachea
Macrophage2

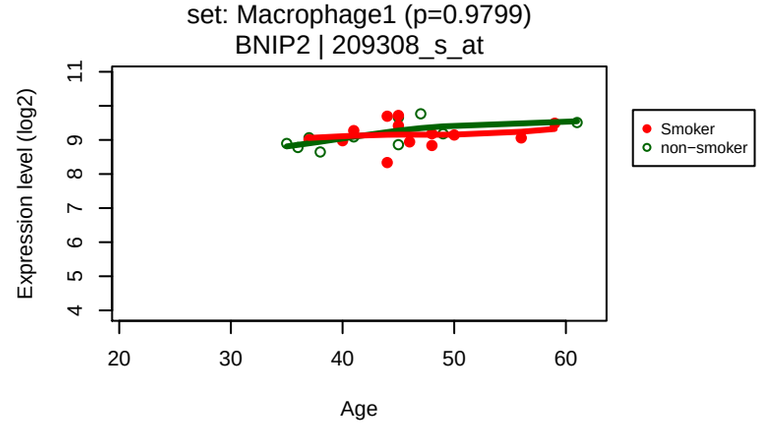

set: SmallAirway1 (p=0.7399)
BNIP2 | 209308_s_at
Expression level (log2)
Age
Smoker
non−smoker

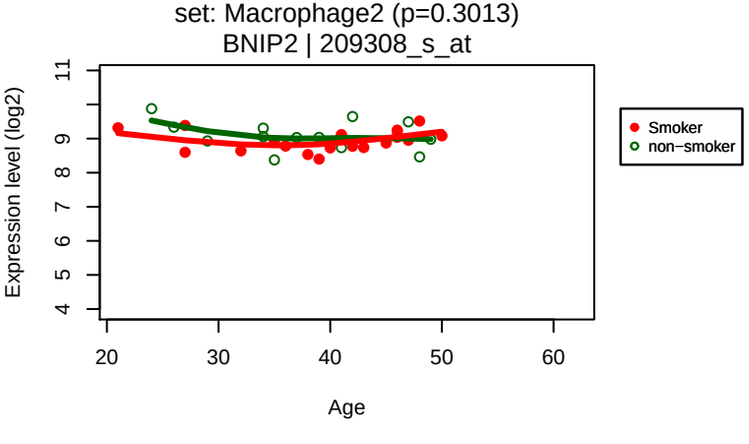

set: SmallAirway2 (p=0.7755)
BNIP2 | 209308_s_at
Expression level (log2)
Age
Smoker
non−smoker

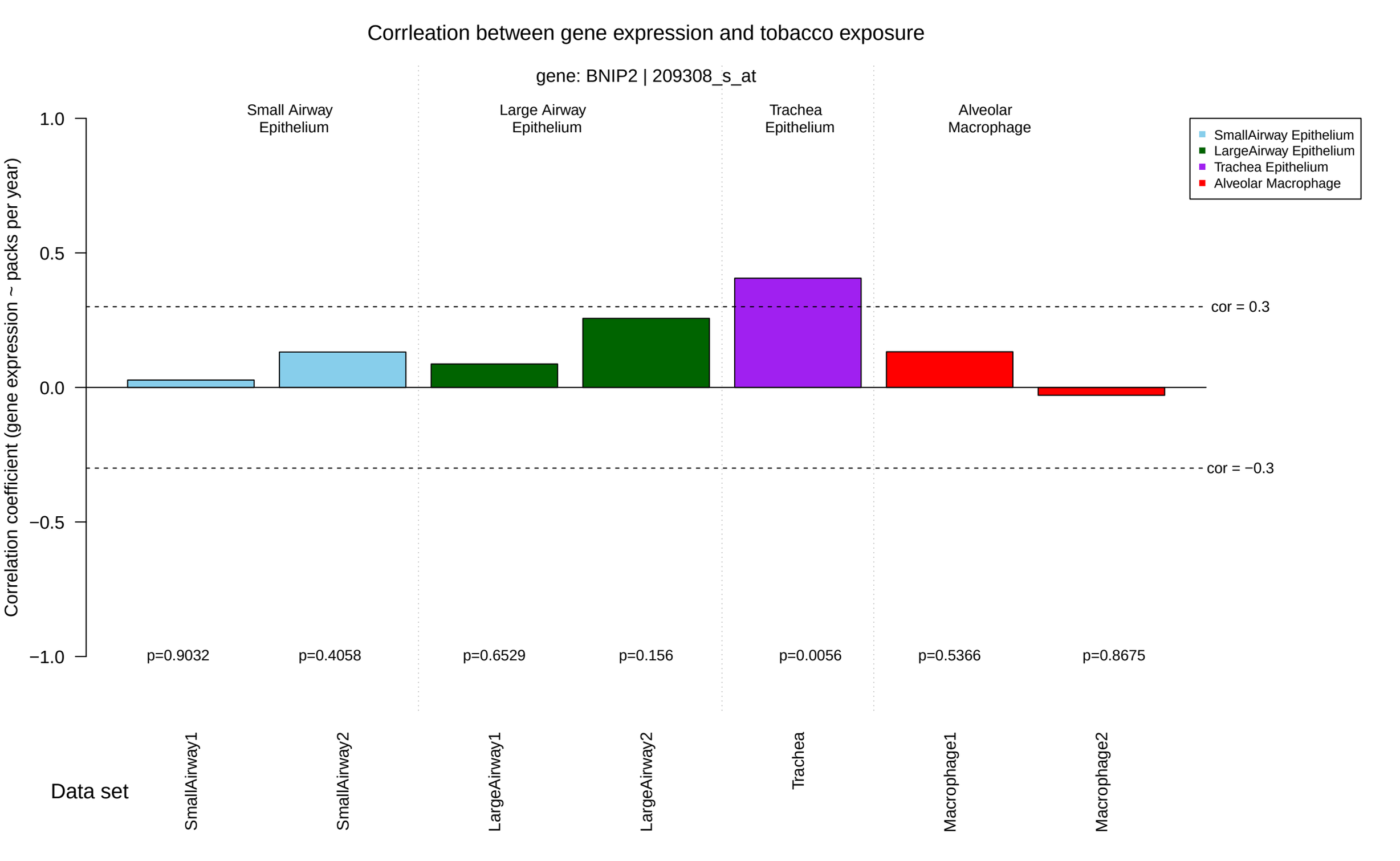

set: LargeAirway1 (p=0.3672)
BNIP2 | 209308_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3575)
BNIP2 | 209308_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0035)
BNIP2 | 209308_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9799)
BNIP2 | 209308_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3013)
BNIP2 | 209308_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BNIP2 | 209308_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9032
p=0.4058
p=0.6529
p=0.156
p=0.0056
p=0.5366
p=0.8675
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2